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$NADH + H^+ + UQ \rightleftharpoons NAD^+ + UQH_2$

**In plants**, an additional NADH dehydrogenase (**External)** complex is present on the outer surface of inner membrane of mitochondria which can oxidise cytosolic NADH + $H^+$.

Ubiquinone (UQ) or Coenzyme Quinone(Co Q) is a small, lipid soluble electron, proton carrier located within the inner membrane of mitochondria.

**2. Complex-II (Succinic dehydrogenase)** It contains FAD flavoprotein is associated with non-heme iron Sulphur (Fe-S) protein. This complex receives electrons and protons from succinate in Krebs cycle and is converted into fumarate and passes to ubiquinone.

Succinate + UQ → Fumarate + $UQH_2$

**3. Complex-III (Cytochrome $bc_1$ complex)** This complex oxidises reduced ubiquinone (ubiquinol) and transfers the electrons through Cytochrome $bc_1$ Complex (Iron Sulphur center $bc_1$ complex) to cytochrome c. Cytochrome c is a small protein attached to the outer surface of inner membrane and act as a mobile carrier to transfer electrons between complex III to complex IV.

$UQH_2 + 2Cyt\ c_{oxidised} \rightleftharpoons UQ + 2Cyt\ c_{reduced} + 2H^+$




Ubiquinone and cytochrome $bc_1$ complex are structurally and functionally similar to plastoquinone and cytochrome $b_6f$ complex respectively in the photosynthetic electron transport chain.

**4. Complex IV (Cytochome c oxidase)**

This complex contains two copper centers (A and B) and cytochromes a and $a_3$. Complex IV is the terminal oxidase and brings about the reduction of 1/2 $O_2$ to $H_2O$.Two protons are needed to form a molecule of $H_2O$ (terminal oxidation).

$2Cyt\ c_{oxidised} + 2H^+ + 1/2\ O_2 \rightleftharpoons 2Cyt\ c_{reduced} + H_2O$

The transfer of electrons from reduced coenzyme NADH to oxygen *via* complexes I to IV is coupled to the synthesis of ATP from ADP and inorganic phosphate (Pi) which is called **Oxidative phosphorylation**. The $F_0F_1$-ATP synthase (also called complex V) consists of $F_0$ and $F_1$. $F_1$ converts ADP and Pi to ATP and is attached to the matrix side of the inner membrane. $F_0$ is present in inner membrane and acts as a channel through which protons come into matrix.

Oxidation of one molecule of NADH + $H^+$ gives rise to 3 molecules of ATP and oxidation of one molecule $FADH_2$ produces 2 molecules of ATP within a mitochondrion. But cytoplasmic NADH + $H^+$ yields only two ATPs through external NADH dehydrogenase. Therefore, two reduced coenzyme (NADH + $H^+$) molecules from glycolysis being extra mitochondrial will yield 2 × 2 = 4 ATP molecules instead of 6 ATPs (Figure 14.10). The Mechanism of mitochondrial ATP synthesis is based on Chemiosmotic hypothesis. According to this theory electron carriers present in the inner mitochondrial membrane allow for the transfer of protons ($H^+$). For the production of single ATP, 3 protons ($H^+$) are needed. The terminal oxidation of external NADH bypasses the first phosphorylation site and hence only two ATP molecules are produced per external NADH oxidised through